5.4 Storing Groceries

The robot stores groceries into a cabinet with shelves. Objects are sorted on the shelves based on similarity, for instance an apple is stored next to other fruits.

Main goal: Move objects from a table to the cabinet, grouping them by category or similarity. Refilling the cereal container.

Optional goals:

1. Opening the cabinet doors
2. Moving a *tiny* object
3. Moving a *heavy* object
4. Picking Objects from the shopping bag

Focus

Object detection and recognition, object feature recognition, object manipulation.

Setup

- **Locations:**
  - **Start location:** Before the test, the robot waits outside the *Arena* and navigates to the testing area when the door is open.
  - **Test location:** The testing area has a cabinet and a table nearby.
- **People:**
  - No people are involved in the test, unless the robot requires human assistance.
- **Furniture:**
  - **Table:** The table has 5–10 objects placed on it and the robot can choose which ones to grasp and in what order. On small tables, objects will be added as the robot frees up space.
  - **Shopping Bag:** There is a shopping bag with 5–10 additional normal items on the ground next to the table.
  - **Cabinet:** The cabinet contains objects arranged in groups — either by category or likeliness — on different shelves.
  - **Cabinet door:** The cabinet is closed. If the robot fails to open the door, it must clearly state this and request the referee to open it.
- **Objects**:
  - **Table objects:** The object on the table are arranged arbitrarily.
  - **Cabinet objects:** The objects are placed behind the cabinet doors and cannot be accessed unless the doors are open.
  - **Containers:** The container for the cornflakes are placed on the table.

Procedure

1. **Table location:** At least two hours before the test, the referees announce the table and cabinet that will be used in the test, as well as a rough location of the table.
2. **Test start:** The robot moves to the testing area when the arena door is open.
3. **Storing groceries:** After identifying the table, the robot moves the objects from the table to the cabinet.
4. **Pouring cereal:** The robot should pour cereal into the designated open container.

Additional rules and remarks

1. **Table:** The table's rough location will be announced beforehand, having its position to the left, right, or behind the robot.
2. **Incorrect categorization:** The score is reduced if an object is stored on the cabinet, but not on a shelf with similar objects; this reduction is applied per incorrectly stored object.
3. **New category:** Objects that do not semantically belong to any of the categories represented on the shelves should be grouped together on a new shelf.
4. **Deus Ex Machina:** The scores are reduced if human assistance is received, in particular for:
   - telling or pointing out to the robot where to place an object
   - moving an object instead of the robot
   - opening the cabinet doors
5. **Communicating Perception**: The robot must clearly communicate its perception to the referee. Pointing at the object or attempting to pick it up is sufficient. When using visualization, make sure the robot tells the referee where to look and make the visualization easily accessible. If the team wants to utilize bounding boxes make sure **only** one object with a bounding box is shown at a time, so the referee is easily able to check and verify. Also make sure the surrounding scene is visible, i.e. just showing a cropped bounding box is not enough.

OC Instructions

At least two hours before the test:

- Announce which table and cabinet will be used in the test.
- Announce a rough location for the table.
- Select which bags will be used in the test.
- Select which container will be used in the test.

Referee Instructions

The referee needs to:

- Place 5–10 objects on the table.
- Place container and cornflakes on the table.
- Place 5–10 objects in a bag near the table.
- Place objects in the cabinet, grouping them by category or likeliness.
- Close the door of the cabinet.

Score sheet

The maximum time for this test is **7:00** minutes.

Action	Score
Main Goal	
Navigating to the table	*15*
Perceiving object and categorizing it correctly	*5×15*
Picking up an object for transportation to the cabinet	*5×50*
Perceiving objects in shelf and saying on which layer the currently handled object should be placed	*5×15*
Placing an object in the cabinet	*5×15*
Placing an object next to similar objects on the cabinet	*5×50*
Opening the first cabinet door	*200*
Opening the second cabinet door	*100*
Pouring cereal into the container	*300*
Bonus Rewards	
Picking up an object from the shopping bag	*5×50*
Picking up a tiny object	*70*
Placing a tiny object	*30*
Picking up a heavy object	*70*
Placing a heavy object	*30*
Autonomously Picking any Object	*50*
Autonomously Placing any Object	*50*
Deus Ex Machina Penalties	
Perceiving object and categorizing it wrongly	*10×−15*
A human handing an object over to the robot	*5×−50*
A human placing an object in the cabinet	*5×−15*
A human placing an object in the cabinet next to similar objects	*5×−50*
A human pointing at a target location	*5×−25*
A human opening the cabinet door	*2×−100*
Spilling cereal while pouring	*−100*
Leaving cereal in the box	*−100*
A human pouring cereal in the bowl	*−300*
Special Penalties & Bonuses	
Not attending (see sec. 3.9.1)	*−500*
Using alternative start signal (see sec. 3.7.8)	*−100*
Outstanding performance (see sec. 3.9.3)	*189*
Total Score (excluding special penalties & standard bonuses)	*1890*